

leet code with my hand easy two sum

1. Two Sum

Easy Topics Companies Hint

Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`.

You may assume that each input would have **exactly one solution**, and you may not use the same element twice.

You can return the answer in any order.

Example 1:

Input: `nums = [2,7,11,15], target = 9`
Output: `[0,1]`
Explanation: Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.

Example 2:

Input: `nums = [3,2,4], target = 6`
Output: `[1,2]`

Example 3:

Input: `nums = [3,3], target = 6`
Output: `[0,1]`

Constraints:

- $2 \leq \text{nums.length} \leq 10^4$
- $-10^9 \leq \text{nums}[i] \leq 10^9$
- $-10^9 \leq \text{target} \leq 10^9$
- Only one valid answer exists.

Follow-up: Can you come up with an algorithm that is less than $O(n^2)$ time complexity?

$< O(n^2)$

```
def twoSum(self, nums: List[int], target: int)
    for i in range(len(nums)):
        for j in range(i+1, len(nums)):
            if nums[i] + nums[j] == target:
                return [i, j]
```

$O(n)$
 $O(n \cdot (n-1))$
 $O(n^2)$

cant for
for x

find two num plus to target